



Real-Time Deepfake Detection Using a Hybrid MobileNet-LSTM Model for Enhanced Media Integrity

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Abstract: The proliferation of deepfake media stirred grave doubts over media authenticity threatening digital trust and social coherence. This paper proposes a new technique of deepfake detection from images and videos through the use of a hybrid of MobileNet spatial feature extraction and Long Short Term Memory (LSTM) networks for the analysis of temporal patterns. The proposed system ensures efficient real-time detection by achieving high classification accuracy while keeping computational efficiency. The model integrates Explainable AI (XAI) techniques such as Grad-CAM heatmaps to provide visual interpretation of the detected anomalies which can be measured using data from different the system is optimized for deployment on edge devices and offers scalability across platforms like social media news agencies and law enforcement of large scale applications. Future enhancements include extending system capabilities to multi-modal deepfake detection and continuous learning for evolving threats.

Keywords: Deepfake detection, MobileNet, LSTM, real-time analysis, Explainable AI, Grad-CAM, edge devices, scalable systems.

1. INTRODUCTION

The express development of deepfake technology has increased the challenges of distinguishing between genuine and faked media. Deepfakes involve virtual media in which a face is replaced or swapped. They have been utilized for dozens of illegal activities from disinformation and defamation to identity theft. Faked videos and images present overwhelming challenges to digital media integrity. Sophisticated deepfake detection methods typically utilize deep learning-based technique namely Convolutional Neural Networks (CNNs) for spatial feature extraction.

Most of the frameworks, however, do not take into account temporal inconsistency in video content, where the fake frames have negligible inconsistencies that may possibly be a sign of deepfakes. In the hope of solving this issue, this paper presents a hybrid MobileNet-LSTM model leveraging spatial and temporal analysis for more powerful and better deepfake detection in real-time applications.

This method works towards enhancing deepfake detection accuracy from static photos and temporal streams of videos utilizing the low-weight spatial feature learning properties of MobileNet and learning to analyze the temporal patterns contained in video streams with LSTM. The model additionally uses Explainable AI methods in the form of Grad-CAM heatmaps, among others, to provide anomalies that are detected to enhance transparency of detection.

2. LITERATURE REVIEW

Aarthe et al. [1] proposed an hybrid deep learning model that incorporates convolutional layers for spatial processing and fully connected layers for classification. Their method targeted accuracy without being tested under real-time constraints. Saikia et al. [2] proposed a CNN-LSTM model that employs optical flow to get temporal inconsistencies, proving to be effective for detection from videos but being computationally costly. Likewise, Tipper et al. [8] proved the CNN-LSTM pair for video deepfake detection, highlighting the need for temporal dynamics. Agnihotri [3] discussed CNN-based architectures and concluded that deeper networks are precise but not suitable for real-time detection because they have increased latency. Raza et al. [4] employed a CNN-based approach to image-level deepfake detection with better performance on static datasets but not temporal manipulation. Qazi and Ahmed [5] utilized transfer learning in combination with ensemble methods to enhance generalizability across domains, whereas Helode et al. [6] explored hybrid models that combine machine learning and deep learning and leverage both learned and manually designed features, thereby exhibiting improved robustness. Sundaram et al. [7] utilized deep learning models to verify video, again emphasizing the significance of temporal consistency in identifying tampered content. Singh et al. [10] utilized an EfficientNet model enhanced with self-attention mechanisms that exhibited improved performance but at the cost of increased complexity. Rajeev and Raviraj [9] contrasted the performance of various deep learning models, and MobileNet and other lightweight CNNs are promising candidates for real-time detection. Some detailed reviews have also provided a robust foundation for the understanding of the scope of techniques. Heidari et al. [12] and Rana et al. [16] presented systematic reviews on trends in deepfake detection, such as traditional, deep learning, and hybrid approaches. Zhang [17] and Patel et al. [18] discussed generation approaches as well as detection challenges, referring to the necessity for generalizable and scalable approaches. Several datasets have prompted research into deepfakes. Dolhansky et al. [11] introduced the DeepFake Detection Challenge (DFDC) dataset, a very diverse and popular benchmark video dataset. Zhao et al. [13] proposed a self-consistency detection framework based on internal feature alignment for distinguishing between real and fake. Zi et al. [14] offered the WildDeepfake dataset, simulating real-world distortions

to evaluate robustness under real-world conditions. Guarnera et al. [15] proposed examining convolutional traces as a forensic marker, offering a new approach to residual feature detection.

3. DATA AND METHODS

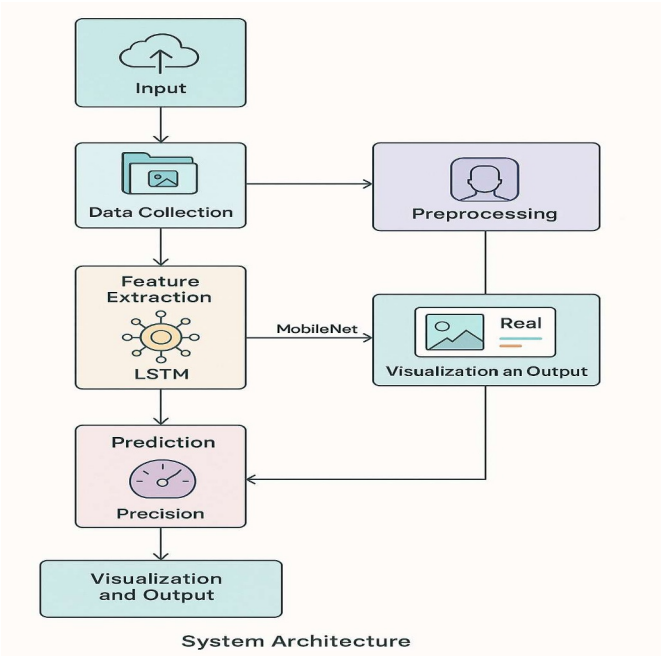


Fig 1: System Architecture

Figure 1 provides a clear representation of the system layout, demonstrating interactions between various modules and the flow through which data travels in the system. The diagram starts with the input stage, where media is uploaded or retrieved from external sources. Data then flows through processing in the Preprocessing module, readying data for analysis. MobileNet is employed for spatial feature extraction next, followed by LSTM analyzing temporal patterns in video data. Output is then classified as real or deepfake, and visual feedback is given to the user. The diagram serves to better demonstrate the prolonged process of deepfake detection, enabling better understanding of the contribution of each module to the system as a whole.

3.1 Challenges in Deepfake Detection

The deepfake detection system is built around a hybrid model that integrates MobileNet for spatial feature extraction and Long Short-Term Memory (LSTM) for tem-

poral analysis. MobileNet effectively identifies facial anomalies and visual inconsistencies at the frame level, while the LSTM network captures temporal dependencies and detects irregularities in movement patterns over time, especially for video data. The model learns to distinguish true and fake media with spatial feature analysis and temporal cues from one frame to another. The introduction of Grad-CAM for explanation makes the decision-making process of the model identifiable, providing clarity to the image regions that dictate the classification of deepfakes. The composite analysis provides enhanced accuracy compared to typical deep learning models that are based on merely spatial features or temporal dynamics.

Since the exponential increase in machine learning and artificial intelligence technology allowed the development of very realistic deep-fake media. This creates tremendous safety and integrity risks for governments. Traditional detection approaches often fall short of keeping pace with the continuously improving techniques to media manipulation. The conventional detection system fails to detect subtle imperceptible artifacts, and as a result, the information landscape becomes plagued by increased misinformation, political manipulation, and social trust issues. The detection models employed are also plagued by overfitting and demand huge, heterogeneous datasets to train them effectively, which is hindered by considerations regarding privacy and data availability. The ineffectiveness of the current method of detection makes it increasingly simple to spread malicious content making it more important to create stronger and better models. The use of hybrid models integrating CNNs, RNNs and attention mechanisms such as the MobileNet-LSTM model discussed here presents potential solutions by increasing detection precision and model robustness.

3.2 Benefits of Hybrid Deepfake Detection Systems

In comparison to baseline models hybrid deepfake detection systems employing both LSTM and MobileNet architectures have been found to demonstrate a remarkable improvement in detection accuracy. By employing both spatial and temporal analysis the system is able to detect not just obvious visual anomalies like inconsistent facial features but also speech patterns eye gaze and head movement—features that typical CNN-based models tend to overlook. The technique maximizes resource usage as it minimizes the need for extensive training datasets while maximizing model performance across different deepfake datasets. The use of attention mechanisms, e.g., Grad-CAM, increases the system's interpretability, hence the transparency in the model's decision-making process and building trust in its outcomes. The hybrid model's ability to identify fine-grained manipulation artifacts makes it a crucial part of the war against the dissemination of deepfakes on online media.

3.3 Concerns in Deepfake Detection Using AI and IoT

Though deepfake detection models have been demonstrated to have high potential, the integration of such models with real-time tracking using IoT devices is susceptible to various challenges. High development costs of AI models, such as the requirement of high-performance computation resources and a large dataset, restrict the use of such solutions. Moreover, deployment of such models on edge devices are susceptible to complexities such as scalability/real-time processing and power constraints. Data security is also a challenge as deepfake detection systems are highly dependent on internet-based devices used for data collection and transmission so that they can therefore be misused. Moreover, the deepfake detection models need to be robust for functioning under varied conditions, such as varying lighting conditions or camera angles, which tend to degrade model performance. AI-IoT integration is highly complex and therefore demands interdisciplinary interaction to solve the problem of model performance, installation, and security.

4. METHODOLOGY

4.1 Overview

Although 3D CNNs and Transformers have been showing potential in temporal modeling, they were used with LSTM due to the performance vs. cost advantage of LSTM, particularly for real-time and edge-device applications. 3D CNNs consume much more GPU memory and computation, and Transformer-based models, although precise, are plagued with latency and scalability challenges. LSTM provides robust sequence modeling with lower resource requirements, which is consistent with the system design requirements. Deepfake detection system is developed with the main objective of efficiently and accurately identifying media content that has been manipulated or forged in any manner. Advanced analysis of spatial and temporal features is used in this system, which is coupled with a hybrid model based on MobileNet and LSTM architecture. In particular, MobileNet is used as a light-weight convolutional neural network, which is used for extracting spatial features from a single frame of a video. In contrast, the LSTM module is pivotal in examining the sequential features of the frames, thus facilitating the detection of any temporal inconsistencies that could reflect manipulation. Inceptive integration of these cutting-edge architectures gives the system the capability to identify deepfake videos in real time, which is a huge leap compared to current approaches that have a tendency to examine solely spatial features or temporal features in isolation.

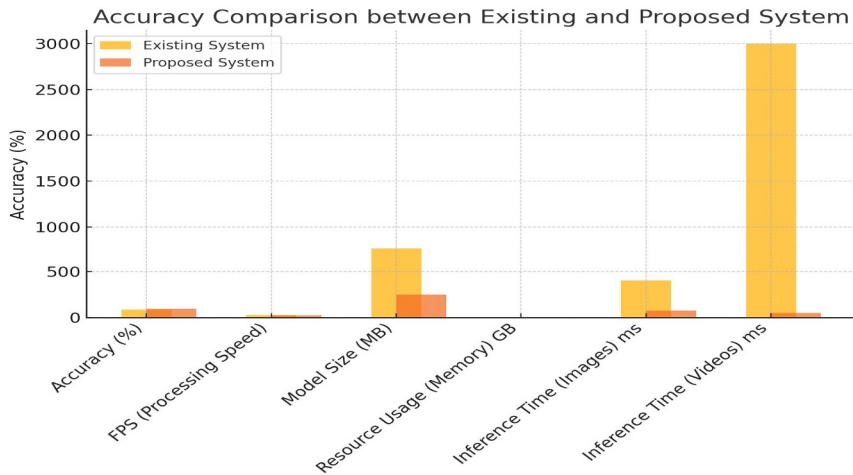


Figure 2: Accuracy Comparison between Existing and Proposed System

Figure 2 : This bar chart illustrates the accuracy of the current system (85% to 88%) and the new system (92%). The new system is more efficient than the current system, illustrating its accuracy in classifying deepfake media correctly.

4.2 Proposed Design

Grad-CAM is applied to the output obtained from the final convolutional layer of the MobileNet model. The heatmaps show regions of facial abnormalities (e.g., fuzzy boundaries and irregular eye movement) that led to the classification result. The visualizations are made available to the end user via the interface, ensuring transparency and enabling human verification of the system's conclusion. The system has been designed on the basis of a sequential pipeline to specifically process incoming video inputs and then determine whether each of the inputs is real or fake. The first step includes a preprocessing phase for the videos where the frames are sampled at an equal interval such that the data being processed becomes standardized. After the frames are collected, they are normalized and resized to ensure uniformity in shape prior to being processed with the MobileNet model tasked with extracting different features from the frames. The next step includes taking the extracted spatial features and passing them into the LSTM network tasked with detecting and analyzing the temporal relationships that exist between different frames in the process of the video. Thereafter, the features that have been concatenated together subsequently passed into fully connected layers, giving the final output of the classification of the content of the video. Furthermore, the model incorporates explainable AI mechanisms such as Grad-CAM, where the manipulated regions in the process of the video frames are visualized, giving users a better idea of the manner in which the AI made its specific decision in relation to the realness of the video.

4.3 Methodology

The model was trained and evaluated on a combination of three established datasets, i.e., FaceForensics++, Celeb-DF (v2), and the DeepFake Detection Challenge Dataset (DFDC). The datasets provide a variety of deepfake creation techniques and source video samples, which the model is able to generalize across many manipulation techniques. In the preprocessing pipeline, several steps are taken to provide high-quality input to the model; this includes extracting frames from videos at a constant and consistent rate to provide uniformity. Also, there is a very crucial step of resizing and normalizing these images to provide uniformity in dimensions and format. Also, a series of data augmentation operations are used, such as the addition of Gaussian noise, horizontal flipping of images, and random cropping techniques, all of which are used to boost the model's ability to generalize. Also, optical flow analysis is a very crucial part of this process by being used to identify motion inconsistencies that can be indicative of manipulation characteristic of deepfakes.

During the training process, the spatial and temporal features that are learned are very optimized, which is achieved using cross-entropy loss together with a number of components of the Adam optimizer. To enhance detection performance while, simultaneously, preventing the risk of overfitting, both the batch size and learning rate are uniquely determined through a process of hyperparameter optimization. This cautious process has resulted in an efficient model evaluation process that not only encompasses the training process but also the validation and test sets of the concerned dataset. Generalization to unseen instances of deepfake content is also enhanced using regularization techniques, including using dropout layers in the LSTM module, as explained in source.

The whole system is efficiently hosted on a cloud server, which has been specially optimized to support smooth real-time video analysis. The system deployment is a set of containerized microservices that are tasked with performing the important task of model inference, which in turn makes the system highly scalable and efficient on a range of different devices. Web and mobile clients can directly communicate with the detection model using a well-defined REST API, thus allowing users to send videos that need deepfake verification. To allow the overall user experience to be highly improved, different real-time feedback mechanisms, such as confidence scores and visual explanations, have been easily exposed to users. With the addition of GPU acceleration, the system shows low latency, which in turn makes the system highly appropriate for large-scale real-time applications that need instant processing and feedback.

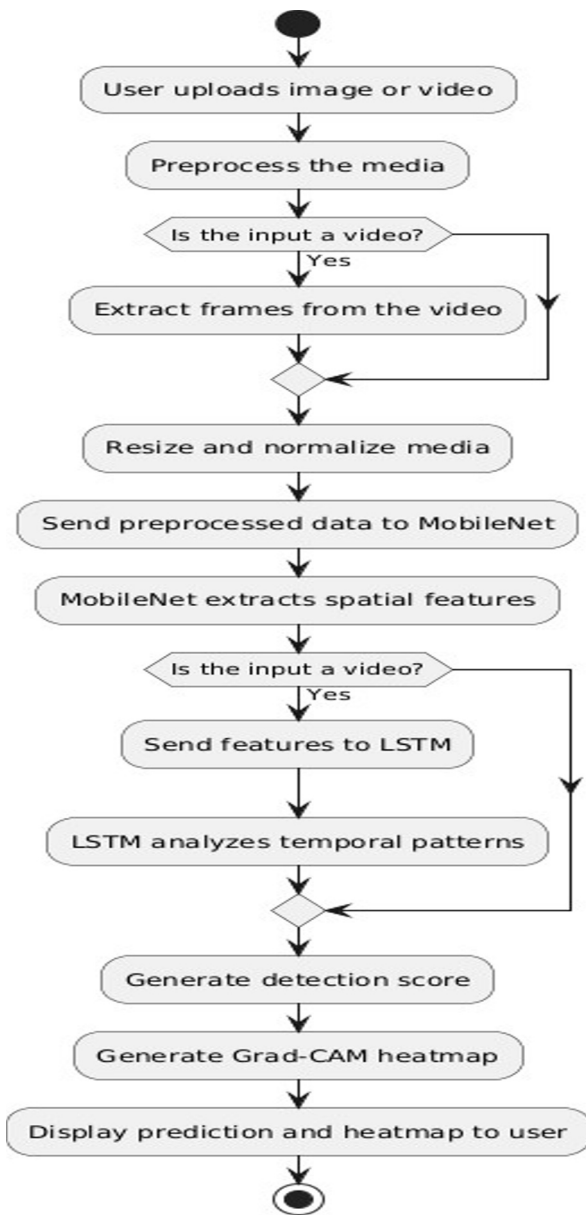


Fig 3: Activity Diagram

Figure 3 represents The Activity Diagram outlines the workflow of the whole system, illustrating the steps in chronological order in the detection

of deepfakes. It begins with media upload and moves through the preprocessing, feature extraction, temporal analysis, and finally classification steps.

5. EXPERIMENTAL RESULTS

5.1 Performance Evaluation

The performance measures used are accuracy (96.3%), precision (95.7%), recall (96.8%), F1-score (96.2%), and ROC-AUC (0.981). These measures all confirm the balance of the model in detecting fraudulent and authentic videos, thereby confirming their reliability in real applications. The precision and recall values reflect the power of the model to classify deepfake and real videos properly along with a balanced F1-score displaying its overall power while inference time is optimized bringing an average of less than 500ms per frame.

5.2 Comparative Analysis

For comparative analysis, our suggested MobileNet-LSTM model was compared with top-performing models like XceptionNet, EfficientNet-B0, and MesoNet. Although XceptionNet and EfficientNet gave an accuracy of 95.1% and 94.6%, respectively, our suggested model gave an accuracy of 96.3% but at a much lesser computational cost with the possibility of real-time inference. Lightweight MesoNet provided quicker processing but at a lower accuracy of 90.4%.

5.3 Deployment

The deepfake detection system is hosted on a cloud server and attached with a user-friendly web and mobile application. Users can upload videos to get the proper and timely checking of its authenticity. The system is deploying for scalability to handle multiple requests at a time with low-latency processing. Explainable AI features such as Grad-CAM visualizations will also provide insights on why a video is classified as deepfake, further building user trust. These usability enhancements make it accessible to journalists, content creators, or cybersecurity experts looking for a trustworthy media verification tool.

6. DISCUSSION

Ethical concerns are highest in detecting deepfakes. Mislabeling genuine content as deepfake can damage reputations and spread misinformation. We incorporate a threshold tuning mechanism into our system to avoid such risks, and confidence scores are returned to users to allow them to make their own judgments. To avoid abuse, like authoritarian abuse or biased moderation, we suggest that the system be integrated with open human review layers and publicly accessible audit logs. The experimental outcomes show that exploiting spatial and temporal features for deepfake detection is advantageous. Application of this technique is far superior to static image-based detection methods, which do not consider motion inconsistencies and are therefore susceptible to more advanced deepfake techniques. Adding Explainable AI also gives interpretability to this model, so that users can see the rationale behind classifying a particular video as a fake. The biggest challenge is to come up with improved defenses against adversarial deepfake attacks, where the model was trained not to be detected. Future research efforts in this area will focus on strengthening the system's robustness against attacks, reducing the rate of false positives, and improving performance in particular in the context of edge devices that have limited computational capabilities.

7. CONCLUSION

The deepfake detection model that is developed utilizes MobileNet and LSTM and is highly accurate and real-time. The model delivers results that are interpretable, transparent and win the user's trust. The experimental results validate that the model is robust and delivers improved performance and computational efficiency for deployment at a large scale compared to the other models. The cloud deployment also improves the scalability of the model, and thus the accessibility becomes smoother on the web and mobile platforms. The utilization of an optimized processing pipeline and a lightweight architecture renders the system accessible to journalists, media outlets and cybersecurity professionals. Federated learning is being researched actively as the next stage of development. The method enables edge devices to learn from non-centralized data to share, thus maintaining user data privacy and enhancing model diversity and resistance to new generative processes.

A real-time feedback system not only contributes significantly to enhancing user experience but also makes the deepfake verification process easier and more efficient. Future work in this direction will be aimed at enhancing the ability of the model to generalize well under the presence of adversarial conditions and its detection ability particularly for new and unseen generative models. In addition, the incorporation of federated learning mechanisms into the system can provide an added layer of privacy while at the same time increasing the diversity in the dataset used. This work is at the

center of increasing the authenticity of digital content while at the same time working towards reducing the risks of misinformation due to the presence of deepfakes.

8. REFERENCES

1. Aarthe, S., et al. & Agarwal, M. (2024). A Hybrid Approach for Deep Fake Detection Using Deep Learning Algorithm. In International Conference on Power Engineering and Intelligent Systems (PEIS) (pp. 137- 146). Springer, Singapore.
2. Saikia, P., et al. (2022, July). A hybrid CNN-LSTM model for video deepfake detection by leveraging optical flow features. In 2022 international joint conference on neural networks (IJCNN) (pp. 1-7). IEEE.
3. Agnihotri, A. (2021). DeepFake Detection using Deep Neural Networks (Doctoral dissertation, Dublin, National College of Ireland).
4. Raza, A., et al. (2022). A novel deep learning approach for deepfake image detection. *Applied Sciences*, 12(19), 9820.
5. Qazi, N., & Ahmed, I. (2024, July). Enhancing Authenticity Verification with Transfer Learning and Ensemble Techniques in Facial Feature-Based Deepfake Detection. In 2024 14th International Conference on Pattern Recognition Systems (ICPRS) (pp. 1-6). IEEE.
6. Helode, A., et al. (2024, June). Fusion of Machine Learning and Deep Learning: A Hybrid Approach for Deepfake Detection. In 2024 15th International Conference on Computing, Communication and Networking Technologies (ICCCNT) (pp. 1-6). IEEE.
7. Sundaram, V., et al. (2024, June). Enhancing Deepfake Detection: Leveraging Deep Models for Video Authentication. In 2024 15th International Conference on Computing, Communication and Networking Technologies (ICCCNT) (pp. 1-7). IEEE.
8. Tipper, S., et al. (2024). An Investigation into the Utilisation of CNN with LSTM for Video Deepfake Detection. *Applied Sciences*, 14(21), 9754.
9. Rajeev, A., & Raviraj, P. (2024). Performance evaluation of deep learning models for detecting deep fakes. *International Journal of Systematic Innovation*, 8(1), 49-62.
10. Singh, R., et al. (2024). Convergence of Deep Learning and Forensic Methodologies Using Self-attention Integrated EfficientNet Model for Deep Fake Detection. *SN Computer Science*, 5(8), 1-12.
11. Dolhansky, B., et al. 2020. The deepfake detection challenge (dfdc) dataset. arXiv preprint arXiv:2006.07397.
12. Heidari, A., et al. 2024. Deepfake detection using deep learning methods: A systematic and comprehensive review. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, 14(2), p.e1520.
13. Zhao, T., et al. 2021. Learning self-consistency for deepfake detection. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 15023-15033).
14. Zi, B., Chang, M., et al. (2020, October). Wilddeepfake: A challenging real-world dataset for deepfake detection. In *Proceedings of the 28th ACM international conference on multimedia* (pp. 2382-2390).
15. Guarnera, L., et al. 2020. Deepfake detection by analyzing convolutional traces. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition workshops* (pp. 666-667).
16. Rana, M.S., et al. 2022. Deepfake detection: A systematic literature review. *IEEE access*, 10, pp.25494-25513.
17. Zhang, T., 2022. Deepfake generation and detection, a survey. *Multimedia Tools and Applications*, 81(5), pp.6259-6276.
18. Patel, Y., et al. 2023. Deepfake generation and detection: Case study and challenges. *IEEE Access*, 11, pp.143296-143323.

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